Technical report — Swiss Statute QA

Summary. A Swiss resident or a front-desk official asks how long the notice period is, when an objection is due, or at what age a right applies, and the answer is fixed by a federal act. Swiss Statute QA contains 537 such facts from 16 federal acts, asked in German, French and Italian (48 also in Romansh); every answer is a verbatim span of the official text in each language and is cited down to the paragraph. Closed-book, Apertus 70B answers 41% correctly and 52% confidently wrong; given the cited article it answers 98%. It knows a fact in all three languages only a third of the time, and on rules amended since 2021 it states the pre-amendment value more often than the current one.

- **Track:** Apertus Readiness — Track 1B (Swiss Voices), core task intelligence
- **Event:** Online
- **Team:** KHLab — Aleksandr Khrukalo
- **Demo:** `make run` (re-grades the recorded responses and checks 32 published numbers); dataset https://huggingface.co/datasets/KHLab/swiss-statute-qa

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Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-8B` and `swiss-ai/Apertus-v1.5-70B` (instruct, as served).
- **How it is used:** evaluation — closed-book and open-book short-answer Q&A, a raw-text cloze probe, and Apertus 70B as the LLM judge.
- **Where it runs:** the hackathon's hosted endpoint (CSCS, OpenAI-compatible); any other endpoint via `LLM_BASE_URL`, `LLM_API_KEY`, `LLM_NAME`.

Use Case Description

A sovereign Swiss model is pitched for public administration and for residents' everyday questions. Most of those questions end at a number in a federal act: the month of notice in OR Art. 335c, the ten days to object to a payment order in SchKG Art. 74, 100,000 signatures in BV Art. 139. A wrong number has a cost (a missed deadline, a lost claim), and a plausible wrong number is worse than "I don't know". The use case is therefore narrow and testable: **does Apertus know what Swiss federal law says, in the language the person asks in, as the law reads today?**

Three facts make this the right moment to measure it. (1) The three language versions of a federal act are equally authentic, so the same question has the same answer in German, French and Italian, which gives a clean test of cross-lingual consistency. (2) Since early 2021 the consolidated acts are published on Fedlex, a JavaScript application whose text sits in a filestore that robots.txt closes to all crawlers (`Disallow: /filestore/*`); the previous publication on admin.ch was open. Apertus filters its web data by robots.txt retroactively (tech report §3.1.1) and did not use its own collected Swiss legal data in pretraining (§H.2). (3) Many resident-facing rules changed after 2021 (AHV 21, the 2023 inheritance and company-law reforms, the 2024 sexual-offence revision, the 2025 civil procedure revision).

Existing Swiss benchmarks do not test this. SwitzerlandQA (Apertus tech report, App. K) is multiple-choice civic knowledge, machine-translated from one source. LEXam tests legal reasoning on law-faculty exams. Neither asks open questions whose answers are verified in each official text, cites the paragraph, or separates amended from unchanged law.

Dataset Summary

Property	Value
Items (facts)	537, one paragraph each, from 16 federal acts
Prompts	1,659: every item in de, fr, it; 48 also in rm
Acts	BV 34, BPR 24, BüG 20, AIG 33, ZGB 50, OR 94, ArG 24, AHVG 32, KVG 21, EOG 19, AVIG 27, SchKG 33, ZPO 35, VwVG 13, StGB 56, SVG 22

Domains	18, e.g. employment 57, criminal law 60, social insurance 56, contracts 46, civil procedure 35, debt enforcement 34
Answer types	duration 330, amount 41, age 39, percentage 39, count 38, fraction 24, majority 5, authority 1, other 20
Amended since 2021	124 items (79 amended, 45 new paragraphs); 90 with a changed answer; 44 carry the superseded answer with verbatim 2021 evidence
Misconceptions	362 items record the typical wrong answer
Difficulty	common 292, specialist 245

Each item holds: the question, a short answer and accepted aliases, and a verbatim evidence span per language; the act, SR number, article, paragraph and a citation (`OR Art. 335c Abs. 1`); a canonical value and unit (`2 month`) for numeric answers; domain, answer type, audience, difficulty, misconception; the temporal label (`unchanged` / `amended` / `new` since 2021-01-01, and whether the change predates March 2024); and the second-pass review verdict.

Data Collection Method

Source. The consolidated acts in force on 1 October 2026, fetched from Fedlex through its public SPARQL endpoint and HTML filestore (`sqa.fedlex`), split into articles and paragraphs, aligned across languages by article id and paragraph position. Two further snapshots support the temporal labels: the version in force on 2021-01-01 and the one in force on 2024-03-01.

Candidates. Paragraphs stating a number, a deadline, a share or a quorum (`sqa.candidates`, 3,965 aligned paragraphs).

Item writing. Items were drafted by an LLM (Claude), one writer per group of acts, following written guidelines (`docs/annotation_guidelines.md`): a question a resident or official would ask, self-contained, no article numbers, one unambiguous short answer, natural wording in each language using that version's own legal term. A separate pass targeted paragraphs whose numbers changed since 2021 and recorded the superseded answer.

Machine checks (`sqa.items`, every item, every language): the evidence is a verbatim span of the cited paragraph; the answer or an alias occurs in it; the canonical value is stated in it (digits or number words, fractions, multipliers, in all four languages); the question does not contain the answer; superseded evidence is verbatim in the 2021 text and states a different value.

Second-pass review (`docs/review_guidelines.md`). A reviewer who did not write the items checked each one against the full article and Swiss law for five properties: unambiguous, current, self-contained, translations equivalent, answer in scope. Of 540 drafts: 476 passed, 61 were repaired (missing exceptions in another paragraph, ordinance overrides, terms renamed by the 2025 ZPO revision, missing superseded answers), 3 were dropped (e.g. AHVG Art. 24 para. 2, no longer applied after *Beeler v. Switzerland*). Every repair was re-run through the machine checks.

Ground truth is the statute text itself; no human-subject data was collected (see Dataset Details).

Dataset Details

- **Licensing.** Swiss federal enactments are not protected by copyright (Art. 5 para. 1 lit. a URG). The questions, answers, labels and model responses are released under CDLA-Permissive-2.0; each item's metadata states the source act, its ELI, the consolidation date and the licence status of the source text.
- **Personal data.** None. No person is named; no data subjects; the only human annotation is the team member's verdicts on judge labels, which contain no personal information.
- **Romansh.** Romansh questions were drafted, not reviewed by a native speaker; answers and evidence are verified in the official Romansh text. They are flagged and reported separately.
- **Known discrepancy.** SVG Art. 68a states 14 days in German, Italian and Romansh and *quinze jours* in French — idiomatically two weeks; no item asks about this deadline.

Evaluation

Method

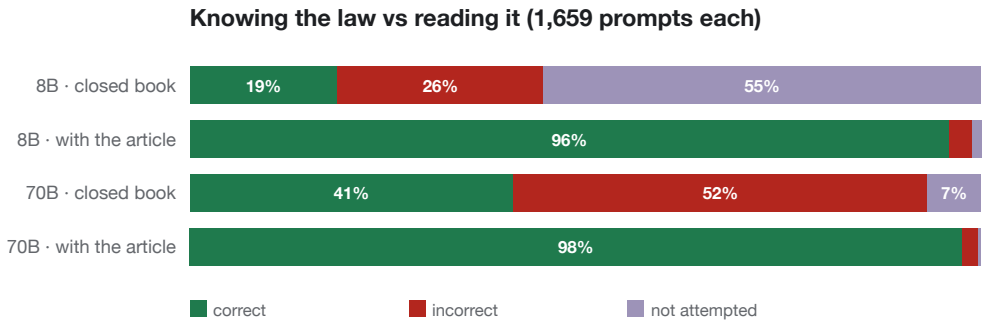
Four conditions per model: **closed book** (question only), **open book** (question plus the cited paragraph in the same language), and a **cloze probe** (the paragraph's own text, with act and article, cut just before the answer, continued by `/v1/completions`). Every answer is graded twice with SimpleQA labels (correct / incorrect / not attempted): by a deterministic **rule grader** (numbers with units in four languages, hedging between values counts as incorrect) and by an **LLM judge** (Apertus 70B, temperature 0, gold answer, aliases and statute evidence in the prompt). A stratified random 20% of the judge verdicts (1,324 of 6,636, seed 20261016) was labelled blind by an independent LLM annotator (Claude, which never saw the judge or rule labels); a page for blind human labelling of the same sample is part of the repository, and human labels are added to the release as they arrive.

Metrics: share correct, not attempted, correct given attempted, F; 95% bootstrap intervals over items; cross-lingual consistency (of items answered correctly in at least one of de/fr/it, the share answered correctly in all three); McNemar tests for paired comparisons; a permutation test that shuffles amendment labels *within each act*, so that a harder act cannot produce the temporal effect.

Evaluation setup and inference parameters

System prompt in the question's language asking for a short answer and permitting "I don't know"; temperature 0; 160 output tokens (cloze: 24); judge 8 tokens. 6,636 answers, 6,636 judge calls, 3,222 cloze calls, all recorded with the exact request in `data/responses/`.

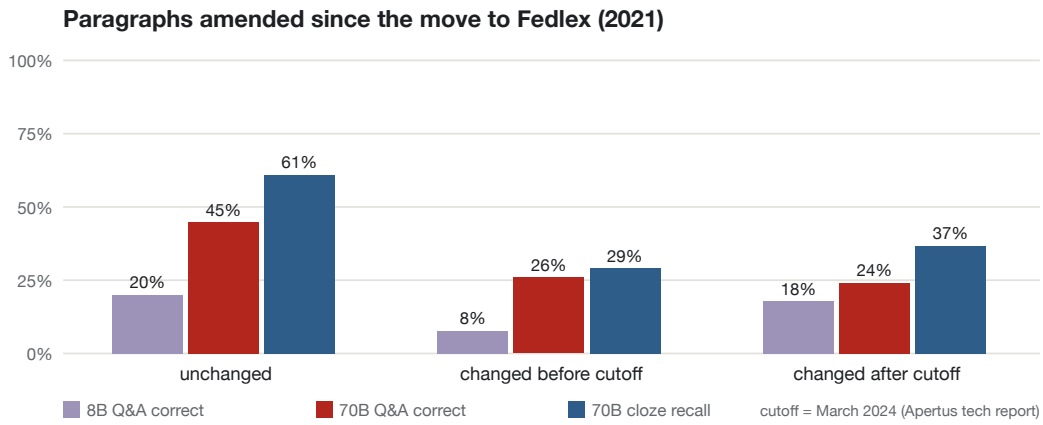
Results



Run	Correct	Not attempted	Correct given attempted	F	Consistency de/fr/it
70B closed	**40.9%** [37.7, 43.9]	6.8%	43.9%	0.424	34% (112 of 330)
70B open	**97.6%**	0.3%	97.9%	0.977	96%
8B closed	18.7% [15.9, 21.4]	55.3%	41.8%	0.258	29% (45 of 155)
8B open	96.0%	1.1%	97.1%	0.966	93%

1. The gap is knowledge, not reading. Given the paragraph, both models answer almost everything (950 prompts flip from wrong to right for 70B, 10 the other way; McNemar $p < 10^{-200}$). Closed-book, 70B is wrong in 52% of prompts and declines in under 7%: when it answers, it is wrong 56% of the time. 8B declines more often (55%) but is no more accurate when it answers (58% wrong).

2. One fact, three answers. Of the 330 facts 70B gets right in at least one language, it gets 112 right in all three. German leads Italian (87 vs 59 discordant items, $p = 0.025$); in German 70B almost never declines (0.7%), in Italian 13%.



3. Amended law is answered with the old law. Within the same act, 70B is 12.3 points less accurate on paragraphs amended between 2021 and March 2024 than on unchanged ones (permutation p = 0.024; 81 vs 269 items in 10 acts) and 26.8 points less on paragraphs amended later (p = 0.0002). On the 101 numeric prompts with a recorded superseded answer, 70B states the **old** value 39 times and the current one 15 times. The cloze probe shows the same from the text side: the current wording of amended paragraphs is completed 29% of the time against 61% for unchanged ones, and given today's text up to the answer, 70B continues with the pre-amendment number 27 times and the current one 11 times. Changes that predate the training cutoff were not learned — consistent with the primary text having left the crawlable web in 2021.

4. Wording without the fact. 70B completes the statute text with the right number in 54% of numeric probes (guessing the most common value per unit: 21%). In 400 probes it completes the text correctly yet answers the same fact wrong when asked a question: the knowledge is there but not reachable by the question a resident would ask.

5. Contamination and source exposure. The answers are spans of public statute text, so the question is not whether Apertus saw the law but which version it saw. We counted Common Crawl captures (index API, 15 crawls 2016–2024, status 200) of each act under its three addresses: the old admin.ch classified compilation, the Fedlex application (a JavaScript single-page app), and the Fedlex filestore that serves the consolidated HTML.

Source (16 acts)	Captures before 2021	Captures 2021–2024	Crawlable for Apertus
admin.ch classified compilation (old text)	724	34	yes (robots.txt allows all)
fedlex.admin.ch application pages	0	566	page is a JavaScript shell
Fedlex filestore (consolidated HTML)	0	111	no: `Disallow: /filestore/*`, and Apertus applies robots.txt retroactively

The text a compliant crawler could read is the pre-2021 admin.ch compilation; the current consolidated text sits behind a robots.txt rule that Apertus's data pipeline honours retroactively, and Apertus did not add Swiss legal corpora to pretraining (tech report, §H.2). This matches result 3: what the model knows is the law as it read before 2021. It also bounds contamination of this benchmark: the current wording of amended paragraphs was not available to the model, and no item, answer or question was published before this release. Limits of the audit: 110 of 720 act × crawl × source cells stayed unanswered after retries (Common Crawl index timeouts, in SVG, VwVG, StGB, AVIG and ArG; they are `null` in `data/results/exposure.json`), and we could not fetch WARC samples to confirm that application captures hold no statute text, because the index was unreachable when we tried (`python -m sqa.exposure_sample` does it when it is up).

Graders. Rule grader and LLM judge agree on 95.2% of 6,636 verdicts (κ = 0.91). On the blind 20% sample the independent annotator agrees with the judge on 97.4% (κ = 0.95; closed book 95.9%, open book 98.8%) and with the rule grader on 94.3% (κ = 0.89). Of the 35 disagreements with the judge, 19 are answers the judge called incorrect and the annotator correct (equivalent units, rounded wording), so the reported accuracies are, if anything, slightly conservative.

Dataset Limitations

- Items are drafted by an LLM and reviewed by an LLM plus a non-lawyer team member; a Swiss lawyer's review would catch subtler scope issues. The two passes changed or removed 64 of 540 drafts, which bounds how many problems remain only loosely.
- Numbers dominate (96% of answers) because they can be verified; normative and procedural knowledge without a number is under-represented.
- Federal law only: cantonal law, ordinances and case law can change what applies in practice; the review removed or narrowed items where it knew of such overrides (e.g. indexed AHV amounts are correct for 2026 and will change on 2027-01-01).
- The 2021 baseline is the consolidation in force on 2021-01-01 (for some acts that is an earlier consolidation, e.g. BPR 2015, because they did not change until later); where Fedlex has no HTML for it, the next HTML consolidation is used.
- The model judge is Apertus 70B itself; the rule grader and an independent LLM annotator are the checks on it. The judge–human agreement the track asks for requires a human labeller; the blind review page is ready, and human labels will be added to `label_human` and reported when collected.

Lessons Learnt and Recommendations

- **Ground Apertus in Fedlex before deploying it for legal questions.** Open-book accuracy is 96–98% for both sizes; retrieval over the current consolidated text closes most of the gap, and the citation in every item makes the retrieval target explicit.
- **Make the authoritative text crawlable, or license it to the model builders.** The federal government's own publication is closed to the crawlers a compliant model may use; the law reaches the model second-hand and out of date.
- **Train abstention on statute questions.** 70B rarely says it does not know (7%) and is wrong more often than right when it answers.
- **Keep the dataset current.** `make live` re-runs any model; re-fetching Fedlex and re-running the checks flags every item whose paragraph changed.

Reproducibility

`make run` builds a Python 3.13 container, validates all 537 items against the statute text, re-grades the 6,636 recorded responses with both graders, scores the cloze probe, recomputes metrics and statistics, and compares 32 headline numbers with `data/results/expected.json` (exit 0 only if all match; about 5 seconds, no network, no GPU, no keys). `make live LLM_BASE_URL=… LLM_API_KEY=… LLM_NAME=…` evaluates another model. `make test` runs the unit tests, each shown to fail on the defect it guards. Seeds: bootstrap 0, permutation 0, human sample 20261016.

License

Code: Apache-2.0. Dataset (questions, answers, labels, model responses): CDLA-Permissive-2.0. Statute text: Swiss federal enactments, not protected by copyright (Art. 5 URG). Report: CC-BY-4.0.

References

Apertus tech report, arXiv:2509.14233 (robots.txt with hindsight §3.1.1; Goldfish loss; Swiss data §H.2; SwitzerlandQA App. K). Wei et al., *Measuring short-form factuality in large language models* (SimpleQA), 2024. Fedlex, Federal Chancellery, fedlex.admin.ch. Swiss Copyright Act (URG), SR 231.1, Art. 5.